

Hybrid anomaly detection

GMM + LSTM-AE Score Fusion with Concept Drift Detection
Advanced Machine Learning · Semester 6

Model A · GMM
F1 90.97% · AUC 95.76%

Model B · LSTM-AE
F1 63.94% → 60.2%

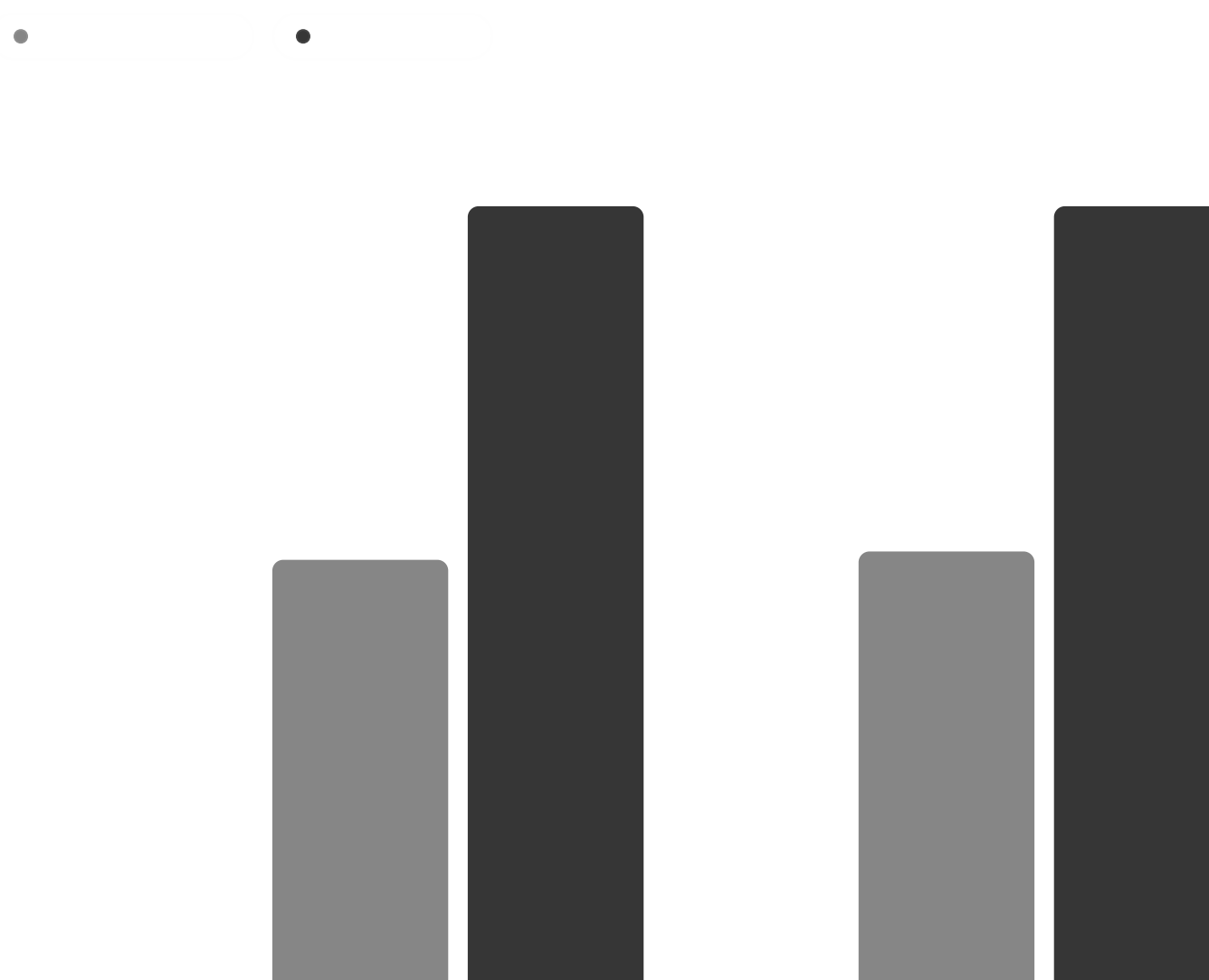
Model C · Hybrid RF ★
F1 93.63% · AUC 98.66%

Evaluation insight

Shuffled windows made LSTM-AE look random — not broken

With 47% attack prevalence and $W=50$, every window contained ~ 23 attacks. Zero benign windows existed — making meaningful AUC impossible. This was an evaluation design failure, not a model failure.

$P(\text{all benign in } W=50) = 0.53^{50} \approx 10^{-14}$ — every window was contaminated



GMM and LSTM-AE see different things

GMM (Model A)

Models marginal distribution $p(x)$

Score: $s_{\text{GMM}} = -\log p(x|\theta)$

No temporal context — memoryless

Strong at: statistically extreme flows (DDoS, Heartbleed)

Weak at: attacks hidden in temporal patterns

LSTM-AE (Model B)

Models conditional $P(x_t / x_{t-1}, \dots, x_{t-W+1})$

Score: mean MSE(window), $W=50$ flows

Strong at: temporal pattern breaks (beaconing, scanning)

Weak at: marginal anomalies (single extreme flow)

Detection coverage by scenario

ScenarioGMMLSTM-AEExample

Marginally anomalous, conditionally normal — **Detects** — **Misses** — Single burst DoS flow

Marginally normal, conditionally anomalous — **Misses** — **Detects** — Bot beaconing

Both anomalous — **Detects** — **Detects** — DDoS flood

Phase 3 complete data flow

Two paths → fusion → decision

GMM PATH



LSTM-AE PATH



Concept Drift Monitor
Page-Hinckley test · $\lambda=50$ · O(1) per flow
Watches GMM log-likelihood stream
Triggers model re-evaluation on drift

Key parameters: GMM — 262 params, O(1) per flow · LSTM-AE — 262,978 params, W=50 windows, latent dim=32 · RF — 100 trees, max depth=5 · Page-Hinckley — $\lambda=50$, O(1) per flow

Three fusion strategies tested

Random Forest outperforms both weighted average and logistic regression by capturing non-linear score interactions

Method A — Weighted average

Formula: $s_C = \alpha \cdot s_{\text{GMM}} + (1-\alpha) \cdot s_{\text{LSTM}}$

Best α : 0.9 — GMM strongly dominant

F1: 69.9% | AUC: 86.2%

Simple linear blend. GMM weight dominates, leaving LSTM signal largely unused.

Method B — Logistic regression

Formula: $s_C = \sigma(w_0 + w_1 \cdot s_{\text{GMM}} + w_2 \cdot s_{\text{LSTM}})$

Weights: $w_1 = 137.45$, $w_2 = 2.26$ (ratio 60.8×)

F1: 90.1% | AUC: 96.1%

Strong improvement over weighted avg. Assumes linear log-odds — misses interaction effects.

Method C — Random Forest ✓ Selected

Config: 100 trees, max_depth=5, balanced classes

Importances: GMM = 0.61, LSTM = 0.39

F1: 93.6% | AUC: 98.7%

Captures non-linear interactions — moderate GMM + high LSTM is more suspicious than either alone.

F1 journey across all phases

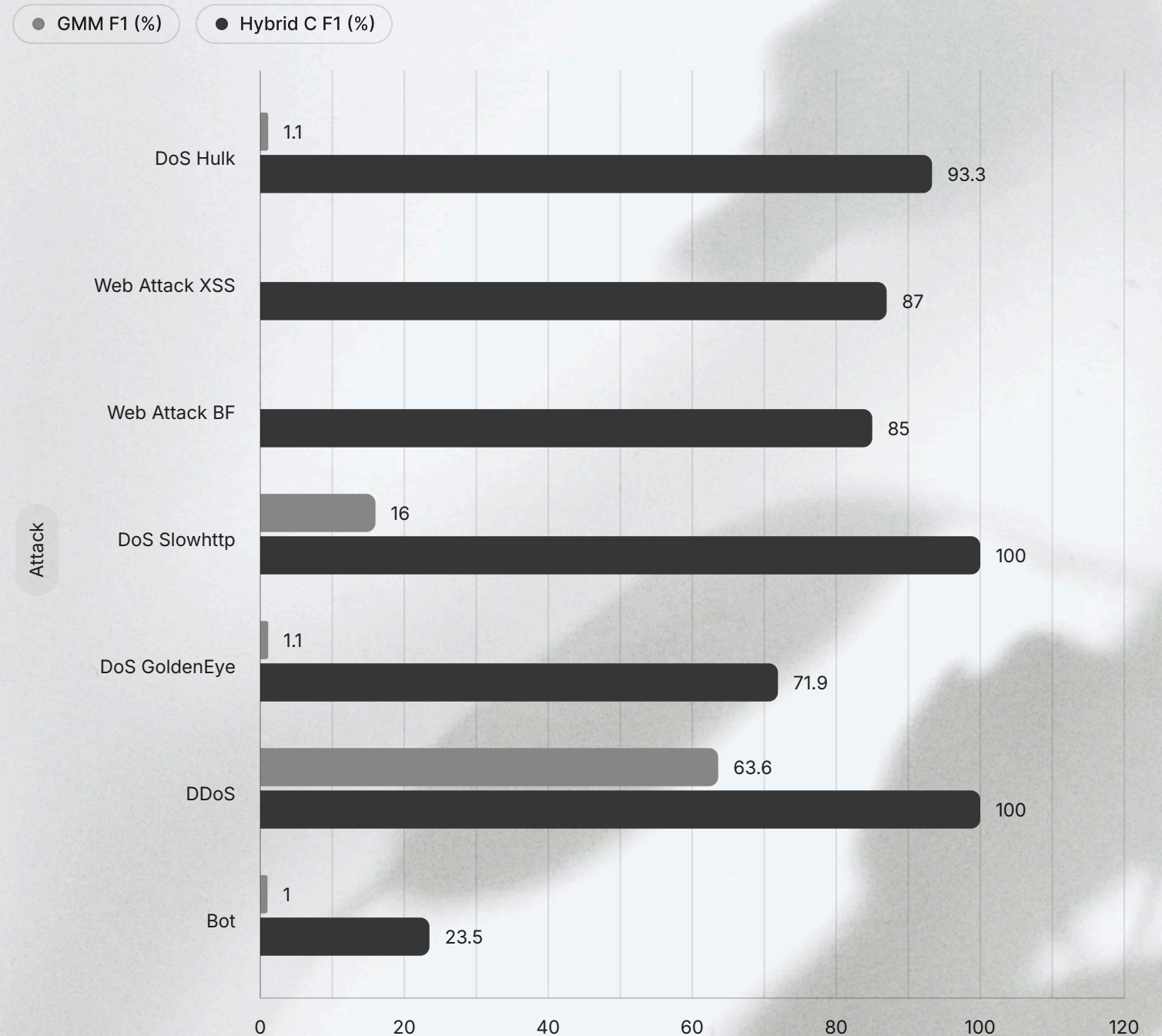
Hybrid RF improves over GMM alone by +2.66pp F1 and +2.90pp AUC — best result across all phases.

Phase	Model	Precision	Recall	F1	AUC
1	Statistical Baseline	50.0%	100%	50.1%	64.8%
1	Isolation Forest	75.4%	51.8%	61.4%	80.3%
1	One-Class SVM	76.8%	74.9%	75.9%	87.2%
1	GMM (Model A)	88.2%	94.0%	90.97%	95.76%
2	LSTM-AE (shuffled)	47.0%	100%	63.94%	52.19% artefact
3	LSTM-AE (temporal)	44.9%	91.5%	60.2%	53.22%
3	Hybrid RF (Model C)	93.2%	94.1%	93.63%	98.66%

Evaluation corrected from Phase 2 to Phase 3, yielding +1pp AUC on LSTM-AE. Phase 2 AUC flagged as artefact of data shuffling.

Attack-by-attack fixes GMM → Hybrid C

Bot remains hard: 1.0% → 23.5%. Beaconing interval exceeds the W=50 window. DoS Hulk's +92.2pp gain reflects RF upweighting the LSTM signal for flows GMM misses on per-flow features alone.



Ablation results

Every component earns its place — remove any one, performance collapses.

	F1	AUC	Drop
Full Hybrid RF (Model C)	93.63%	98.66%	—
GMM only (remove LSTM)	74.4%	95.76%	−19.2pp
LSTM only (remove GMM)	60.2%	53.22%	−33.4pp
Equal weight ($\alpha=0.5$, no RF fusion)	63.9%	56.75%	−29.7pp

Key takeaways: GMM is the dominant component — removing it costs 33.4pp F1. But LSTM still adds genuine signal: dropping it loses 19.2pp. Equal weighting alone collapses to Phase 2 levels, proving RF fusion is the critical differentiator.

Drift detection

Page-Hinckley test detects regime shifts at line-rate speed

$PH_t = \max(\bar{\mu}_k) - \bar{\mu}_t$ on GMM log-likelihoods.
Baseline mean LL = -28.6 (5,000 benign flows).
 $O(1)$ per flow — suitable for real-time monitoring.

$\lambda = 50 \rightarrow 1$ drift event, detected 13 flows into FTP-Patator attack



Summary

F1 journey

50.1% Base → 90.97% GMM →
60.2% LSTM (temporal) → 93.63%

Hybrid C

What worked

Temporal ordering restored LSTM-AE's benign baseline, eliminating the shuffle artefact.

RF fusion captured non-linear score interactions, gaining +29.7pp over equal-weight blending.

Page-Hinckley drift detector identifies the benign-to-attack boundary in real time.

Still hard

Bot (23.5%): beaconing interval exceeds the $W=50$ window.

HTTP-level attacks: flow stats alone are insufficient — payload inspection needed.

Future work

Longer windows ($W=500$) for slow beaconing.

Session-level graph features: IP contact frequency.

Online LSTM-AE fine-tuning on confirmed-benign drift windows.

Adversarial retraining protection against poisoned drift triggers.

Multi-class Model C to identify attack family, not just binary label.

